Jiaqi Ge

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EDUCATION

Guangdong Technion – Israel Institute of Technology Sep 2023 – Present
B.Sc. in Mathematics with Computer Science | English-taught Program

Technion – Israel Institute of Technology, Haifa, Israel Summer 2026
Summer Research Exchange in Non-Euclidean Deep Learning

PUBLICATIONS

Manuscript on Repository Knowledge Acquisition for Software Issue Resolution
Co-author | Under submission

RESEARCH EXPERIENCE

Research Exchange Student – Fast Document Classification with FSW Summer 2026
Nadav Dym's Research Group | Technion – Israel Institute of Technology | Haifa, Israel | supervised by Tal Amir

- Collaborated on an optimal-transport document-learning pipeline that represented texts as probability measures over 300-dimensional fastText word embeddings and mapped them into fixed-dimensional vectors using Fourier Sliced-Wasserstein (FSW) embeddings.
- Benchmarked FSW-based kNN against exact Word Mover's Distance on Amazon, Classic, and Reuters datasets, reducing per-query classification time by approximately 90–155×; developed a shortlist-and-rerank strategy that preserved near-WMD accuracy with 4–29× speedups.
- Extended the Euclidean representations to k-means and spectral clustering, UMAP visualization, and supervised MLP/contrastive learning; the trained models matched or outperformed exact WMD on two of three benchmarks.

Research Assistant – Repository QA for Software Issue Resolution Feb 2026 – Present
Shanghai Jiao Tong University | advised by Prof. Xiaodong Gu

- Contributed to manuscript writing and revision, including related work, experimental results, method analysis, ablation discussion, and figure/result interpretation.
- Designed and executed baseline vs. QA-augmented experiments for repository-level software issue resolution.
- Built a judge-based evaluation pipeline for question-level, set-level, and pairwise assessment, and verified the scoring and aggregation process.
- Conducted fail-to-pass trajectory analysis, including repair-stage segmentation, QA-usage analysis, and patch-alignment analysis, and prepared supporting figures and reports.

Research Assistant – Mixture-of-Experts (MoE) Apr 10, 2026 – Present
MBZUAI – Mohamed bin Zayed University of Artificial Intelligence | advised by Prof. Lijie Hu

- Conducting research on Mixture-of-Experts (MoE).

Research Assistant – Metamorphic Testing for Explainable AI Oct 2025 – Feb 2026
Guangdong Technion – Israel Institute of Technology | advised by Prof. Nazareno Aguirre | Project Score: 90/100

- Built a grammar-based fuzzing pipeline to generate multi-step text perturbation chains (typos, punctuation, entity swaps, negation, ...) using Checklist.
- Implemented candidate metamorphic-property generation and checking (INV invariance / DIR directional) to evaluate model behavior under k-step transformations.
- Integrated a HuggingFace sentiment-analysis model with Checklist to run automated INV/DIR test suites at scale, and summarized experimental findings for research documentation.

Research Assistant – Numerical Methods for Boundary Value Problems Aug 2025 – Nov 2025
Guangdong Technion – Israel Institute of Technology | advised by Prof. Antti Rasila | Project Score: 95/100

- Conducted research on numerical solutions for boundary value problems (BVPs) in MATLAB.
- Reviewed classical theory including existence/uniqueness, Green's functions, and Sturm–Liouville problems.
- Compared shooting, finite-difference, and spectral methods for linear and nonlinear BVPs.
- Used the MATLAB Chebfun toolbox to solve nonlinear systems, singular Sturm–Liouville problems, and chaotic dynamical systems.
- Authored a 20-page report titled "Solving Boundary Value Problems Using Chebfun".

WORK EXPERIENCE

Calculus II Homework Grader Mar 2026 – Aug 2026
Guangdong Technion – Israel Institute of Technology | Shantou, China

- Graded weekly Calculus II homework using course rubrics, evaluating mathematical correctness, reasoning, and clarity of solutions.
- Provided consistent written feedback and documented recurring errors to ensure fair assessment and help students improve.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant – Linear Algebra & Calculus Mar 2026 – Aug 2026
Guangdong Technion – Israel Institute of Technology | Shantou, China

- Facilitate student learning in weekly workshops by providing guidance on complex mathematical problems and theorems.
- Assist students in mastering foundational concepts of Linear Algebra and Calculus through interactive problem-solving sessions.

Undergraduate Teaching Assistant – Python Oct 2025 – Jan 2026
Guangdong Technion – Israel Institute of Technology | Shantou, China

- Assisted in teaching the introductory Python course and supervised lab sessions.
- Supported students with debugging code and understanding core programming concepts.

AWARDS

Chancellor's Scholarship – 28,500 US Dollars in total 2023 – 2027
Awarded by Guangdong Technion – Israel Institute of Technology for outstanding performance in the Gaokao (Chinese National College Entrance Examination).

SKILLS

Programming: Python, C, C++, Java, MATLAB, Assembly
Languages: Chinese (Native), English (Fluent)